

KYOTO, Japan

WMO: 477590

Lat: 35.0147N

Lon: 135.7328E

Elev: 173

StdP: 14.60

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.5	32.0	15.9	12.4	40.4	19.2	14.5	40.9	13.0	41.7	11.8	42.6	3.7	260	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.9	95.5	76.6	93.4	76.1	91.3	75.6	78.6	90.2	77.7	88.9	76.9	87.6	7.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	134.6	83.1	74.6	130.5	83.1	73.8	126.8	82.9	42.2	90.7	41.3	89.1	40.5	87.7	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.3	11.2	9.8	DB	27.2	98.6	2.0	1.9	25.8	100.0	24.7	101.1	23.5	102.2	22.1	103.6
			WB	24.4	79.8	2.4	1.2	22.6	80.7	21.2	81.4	19.8	82.0	18.0	82.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	41.0	42.3	48.2	58.3	67.7	74.4	82.2	84.5	76.9	66.1	54.9	45.3	(6)
	DBStd	15.77	3.79	4.86	5.73	6.19	4.70	4.24	4.40	3.55	5.16	5.13	5.45	4.83	(7)
	HDD50	793	280	223	106	9	0	0	0	0	0	0	15	160	(8)
	HDD65	3102	744	636	521	214	26	0	0	0	1	47	303	610	(9)
	CDD50	5144	1	7	49	257	550	732	997	1069	806	499	163	14	(10)
	CDD65	1979	0	0	0	12	111	282	532	604	356	81	1	0	(11)
	CDH74	18828	0	0	1	93	720	1905	5736	7200	2848	323	2	0	(12)
	CDH80	7527	0	0	0	8	149	540	2470	3386	937	37	0	0	(13)
(14) Wind	WSAvg	4.2	3.8	4.0	4.5	4.6	4.7	4.4	4.4	4.7	4.3	3.7	3.4	3.6	(14)
(15) Precipitation	PrecAvg	62.8	3.5	3.6	4.5	4.4	6.0	7.3	7.8	5.6	7.4	5.5	3.9	3.4	(15)
	PrecMax	80.2	6.4	5.6	6.9	7.3	16.5	18.1	15.1	18.3	16.1	14.6	7.7	5.0	(16)
	PrecMin	40.9	1.5	1.6	1.6	1.1	1.7	2.6	2.0	1.8	2.3	0.9	0.7	1.3	(17)
	PrecStd	10.0	1.2	1.1	1.4	1.8	3.3	3.2	4.1	3.8	3.7	3.0	1.9	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	63.2	70.8	80.9	87.9	91.4	97.8	98.9	93.6	83.7	72.8	62.7	(19)
		MCWB	49.4	52.4	55.0	60.3	65.0	72.8	77.5	76.8	75.3	70.1	60.3	55.2	(20)
	2%	DB	52.9	57.7	66.0	76.6	83.9	88.0	95.2	96.2	90.6	80.2	69.4	59.0	(21)
		MCWB	45.2	48.6	53.1	58.6	63.8	70.7	76.7	76.6	74.4	67.3	58.9	51.2	(22)
	5%	DB	50.5	54.2	62.1	73.1	81.0	85.7	93.0	94.3	88.0	77.5	66.7	56.4	(23)
		MCWB	43.3	45.8	50.9	57.1	62.7	69.7	76.2	76.4	73.4	65.3	57.7	49.0	(24)
	10%	DB	48.2	51.2	58.9	69.6	78.2	83.1	90.7	92.3	85.5	75.0	64.1	53.8	(25)
		MCWB	41.7	44.2	49.6	56.3	61.8	69.0	75.7	76.1	72.1	64.0	56.3	47.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	54.9	58.8	67.1	71.3	76.7	79.7	79.9	77.6	73.5	64.3	57.8	(27)
		MCDB	55.4	59.7	63.3	72.0	78.6	86.0	93.1	92.3	87.2	79.9	67.2	60.5	(28)
	2%	WB	47.1	50.8	55.9	63.1	68.6	75.1	78.3	78.9	76.4	70.4	61.8	53.1	(29)
		MCDB	50.8	55.7	62.5	70.4	76.2	82.9	90.2	90.6	85.9	77.1	66.3	57.4	(30)
	5%	WB	44.4	47.6	53.3	60.6	66.4	73.6	77.4	78.1	75.4	68.0	59.7	50.1	(31)
		MCDB	49.3	52.3	59.8	68.9	74.4	81.1	88.5	89.5	84.3	74.5	65.1	54.8	(32)
	10%	WB	42.1	44.7	50.6	58.4	64.5	71.9	76.6	77.4	74.4	65.8	57.6	47.6	(33)
		MCDB	47.3	50.6	57.6	66.3	74.1	79.4	87.0	88.4	82.7	72.8	63.3	53.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	13.6	16.2	18.0	17.7	14.6	13.9	14.9	13.9	14.8	14.6	13.2	(35)
		MCDBR	15.8	18.8	21.7	23.6	23.1	18.8	17.2	17.4	16.4	17.2	17.0	16.2	(36)
		MCWBR	9.8	11.5	12.5	10.8	9.7	7.2	5.4	5.3	5.8	7.7	9.3	10.0	(37)
	5% WB	MCDBR	13.7	15.5	17.0	16.4	16.0	13.2	15.3	15.1	13.6	13.0	13.4	13.8	(38)
		MCWBR	10.1	11.8	12.5	10.2	9.4	6.1	5.2	5.0	5.5	7.4	9.4	10.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.434	0.474	0.541	0.573	0.577	0.599	0.598	0.562	0.510	0.469	0.441	0.427	(40)
	taud		2.025	1.940	1.807	1.787	1.835	1.848	1.902	1.982	2.087	2.113	2.089	2.059	(41)
	Ebn at Noon		231	237	232	233	234	229	228	233	239	237	228	224	(42)
	Edh at Noon		43	52	64	69	66	65	62	56	48	43	40	39	(43)
(44) All-Sky Solar Radiation	RadAvg		661	868	1196	1484	1634	1490	1553	1598	1256	1010	751	601	(44)
	RadStd		48	77	96	133	171	136	219	155	108	97	63	44	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page